

Welcome to the 2006 Psychology and Neuroscience In-House Convention

Thanks for attending the 32nd Annual In-House Convention.

Thanks should also be extended to the In-House Convention Committee for planning and organization. The committee consists of the following individuals: Sophie Jacques, Angela Arsenault, Sabrina Bellhouse, Martha Carmichael, Natalia Jaworska, Kerry MacSwain, and Tim O'Leary.

Friendly Reminders

- * All talks are in LSC 4258/4263.
- * Please bring your own mug for coffee or tea!
- * All presentations are listed in the name of the presenter from the abstracts submitted.
- * There will be a reception at the end of Friday in the 2nd Floor Lounge.

Thursday, May 4, 2006

- 9:00 Richard Brown: Introduction and brief history of convention
- 9:30-10:30 **Session 1.** *Chair:* Natalia Jaworska
T2) Lisa Boutilier, Daniel Clark, Faith Conrad, Julie Longard, Sarbjeet Singh
T3) Ron Hoffman
- 10:30-11:00 **Coffee Break**
- 11:00-12:15 **Session 2.** *Chair:* Timothy O'Leary
T4) Erin Mazerolle
T5) Jodie Gawryluk
T6) Martha Carmichael
T7) Beverly Butler
T8) Mike Lawrence
- 12:15-1:45 **Lunch Break and Posters** (Psychology 2nd Floor Lounge)
P1) Tessa Craig
P2) Jennifer Parker
P3) Meredith Pike
P4) Laura Knickle
- 1:45-3:00 **Session 3.** *Chair:* Bradley Frankland
T9) Amy MacPherson
T10) Chris Moore
T11) Joshua Salmon
T12) Darren McKee
T13) Richard Wassersug
- 3:00-3:30 **Coffee Break**
- 3:30-5:00 **Session 4.** *Chair:* Sabrina Bellhouse
T14) Nancy Bandstra
T15) Steve Porter
T16) Kerry Lynn MacSwain
T17) Sherry Stewart
T18) Chris Mushquash
T19) Shannon Currie
- 5:00 Reception in the 2nd Floor Lounge

Friday, May 5, 2006

9:15-10:30 **Session 5.** *Chair:* Martha Carmichael
T20) Alana Paul
T21) Matthew Kutcher
T22) Donald Mitchell
T23) Simon Gadbois
T24) Oliver Braubach

10:30-11:00 **Coffee Break**

11:00-12:30 **Session 6.** *Chair:* Kerry MacSwain
T25) Jennifer Vriend
T26) Esther Lau
T27) Sabrina Bellhouse
T28) Lindsay Uman
T29) Angela Arsenault
T30) Jonathan Fawcett

12:30-2:00 **Lunch Break and Posters** (Psychology 2nd Floor Lounge)
P5) Sarah Ironside
P6) Adrienne Junek
P7) Erin Marzerolle
P8) Jodie Gawryluk

2:00-3:00 **Session 7.** *Chair:* Angela Arsenault
T31) Timothy O'Leary
T32) Lisa Wright
T33) Gillian Alcolado
T34) Sarah Boudreau

3:00-3:30 **Coffee Break**

3:30-5:00 **Session 8. *BRUCE MOORE MEMORIAL SYMPOSIUM*** *Chair:* Sophie Jacques
T35) Richard Brown - The life and Work of Bruce Moore at Dalhousie
T36) Vin Lolordo - Bruce Moore's Contributions to Conditioning
T37) John Christie - Doves in Space
T38) Trish Cole - Parrots and Mynahs and Crows, Oh My!
T39) Chris Moore - Bruce Moore's Contributions to the Phylogeny of Learning

5:00 Closing Comments & Awards Recognition

5:15 Dinner/Reception in the 2nd Floor Lounge

T2) The History of Psychology at Dalhousie

*Lisa Boutilier, Daniel Clark, Julie Longard, Sarbjeet Singh
(with research by the 2005-6 class of Psychology 3580)*

Psychology at Dalhousie has a rich past, filled with interesting personalities, and dedicated and driven researchers. In some ways, it is a microcosm of the development of disciplinary psychology over the past century and a half, yet with a unique Nova Scotian flavour. This is the story of an unwelcome Freudian and a Prince whose connections to the President allowed him to move the department into the White House. It is also the story of an award-winning architectural wonder and how respectability was brought to “snake oil salesmen.” And it is the story of a chairman who brought wolves to Nova Scotia, and another who added a seal tank to the LSC. And it is also a lament that Richard Brown’s exile to Sable Island failed. The class interviewed past and present faculty and staff, including six department chairs. Students conducted extensive archival research, and began the construction of an extensive report on the development of this leading psychology department in Canada.

T3) We've Come a Long Way in 35 years

Ron Hoffman

In September of 1971, the Life Sciences Centre opened to house the departments of Psychology, Biology and Oceanography (in order of importance, although you could argue the last two) at Dalhousie. Just four years earlier, the Tupper Medical building had opened, complete with basement television studio, bringing home-grown moving imagery to modern teaching at the university. The Psychology Department enthusiastically embraced this new technology for both teaching and research, and pushed to have a television studio included in the new LSC. The first consumer-level video recording equipment appeared at the same time, and our initial occupation of the LSC was accompanied by much video making. Most of that early work has been lost, and much of what remains is nearly irredeemable due to deterioration of the media and changing standards in video technology. This presentation will give you a taste of the historical records we are losing and suggest that perhaps we should be doing a better job of documenting the history of what arguably has been one of the most influential Psychology departments in Canada.

**T4) Determining neuro-cognitive connectivity: An event-related fMRI study at 4T. OR
Beyond phrenology: Functional connectivity of the human brain**

Erin L. Mazerolle and Ryan C.N. D'Arcy

There is a growing trend in neuroscience to study the nature of the connections between brain regions, as these connections are vitally important to understanding brain function. The well-studied visual and motor systems provide an excellent foundation for investigations of functional connectivity. Interhemispheric connections between the visual and motor systems have been previously studied using unimanual reaction times (RTs) to lateralized light flashes. When the light is flashed in the hemifield contralateral to the responding hand, the visual information must cross to the hemisphere that controls the hand. While behavioural studies have yielded differences between crossed and uncrossed RTs, it is likely that light flashes can be processed by relatively low-level systems. We propose to use lateralized cognitive processes (face and word processing) to elicit interhemispheric transfer between distributed cortical systems. Furthermore, we plan to vary the extent of the network recruited by the task by manipulating the number of interhemispheric transfers. This model will allow for testing of basic hypotheses about 'neuro-cognitive' connectivity using non-invasive functional imaging.

T5) Is emotional intelligence affected by right more than left temporal lobe resections?

Jodie Gawryluk, Jeannette McGlone, Mark Sadler, and Ryan D'Arcy

The right amygdala may have a distinct role in emotional intelligence. This research will examine emotional intelligence after left and right temporal lobe resections that include the amygdala. There are several approaches to evaluate emotional intelligence. Critically, this study will use Bar-On's Emotional Quotient inventory. It is hypothesized that individuals with right temporal lobe resections will score lower on the EQ-i, than those with left temporal lobe resections. Conversely, individuals who have had left temporal lobe resections are expected to have lower verbal memory and naming scores than those with right-sided resections. These groups are not expected to differ on descriptive information or neuropsychological measures such as intelligence, speed of reading, or visuospatial skills.

T6) Ear asymmetries in auditory temporal gap detection

Martha E. Carmichael and Dennis P. Phillips

There is a long history of interest in the functional specializations of the two cerebral hemispheres in man. In particular, there is interest in the reasons why language processing is often lateralized to the left hemisphere. One hypothesis is that the left hemisphere, and perhaps especially the auditory part of it, is in some way a "faster temporal processor" than is the right. The intended connotations of that expression are not always clear, but they typically include auditory temporal acuity, i.e., the temporal

"grain" with which the stream of sound is perceptually resolved. In this regard, high temporal acuity is advantageous for resolving the sounds of speech. The present study employs temporal gap detection paradigms, with and without two kinds of contralateral masking, to measure temporal acuity through the left and right ears in normal adult listeners. Differences in acuity through the two ears may be informative about the temporal resolution supported by the two cerebral hemispheres.

T7) Orienting attention in visuospatial neglect: How limb movements affect the disengage deficit
Or

"Limbs...Cueing...Attention" (read as "Lights...Camera...Action")

Beverly Butler

Visuo-spatial neglect from right hemisphere damage is associated a difficulty disengaging attention from the right side to move it to the left side (disengage deficit). Neglect has been shown to be reduced with both active and passive left limb movements, but the mechanisms of this positive effect is unknown. In the current study a cuing paradigm was used to assess the effect of active (AM) and passive (PM) left and right limb movements, compared to a no movement condition (NM), on the disengage function in right hemisphere stroke patients with and without neglect and in older healthy controls. Results showed that the disengage deficit was related in size to neglect severity in stroke participants and orienting was affected by limb movements in the older control group and in two neglect subjects with a severe disengage deficit. These data suggest that left limb movements can directly affect the orienting of attention, although the direction and consistency of the effect remains yet to be resolved.

T8) Patterns of attention in visual search after stroke

Mike Lawrence, Beverly Butler and Gail Eskes

Visual search performance was measured in healthy subjects and right hemisphere stroke survivors, including patients with and without neglect. These groups were compared on traditional measures of global and spatial attention, as well as new measures that quantified how attention moved through space during the search task. It was discovered that neglectful stroke survivors fared worse than their non-neglectful and healthy counterparts not only in where they allotted their attention, but how they moved it as well. These results reveal a novel pattern of performance that may shed light on the mechanism of the neglect dysfunction.

T9) Individual Differences in the Development of Face Recognition

Amy C. MacPherson, Patricia A. McMullen, and Phillip J. Dunham

Children at seven (n = 32) and eleven (n = 30) years of age participated in a project examining individual differences in face recognition and the processing strategies children use to recognize

faces. Children took part in two tasks including choosing the familiar face from a series of upright or inverted photographs of human faces and determining which of two abstract images (i.e., Mooney face stimuli and scrambled counterparts) contained a face. The goal of this project was to provide further evidence of a switch during childhood from feature-based to configural processing in the recognition of upright faces. Results are combined with those from adults ($n = 62$) tested under similar conditions to tell a complete developmental story.

T10) Studies in visual-proprioceptive integration

Chris Moore

Integration of information over visual and proprioceptive modalities is an important means for establishing a sense of self in early development. Recently, we have been studying this process in adults through two approaches. First, we examine the temporal parameters of visual-proprioceptive integration by showing adult participants delayed video (from 0 to 5 video frames) of their own arm and facial movements. Our results show a discrimination of live from delayed video for both hand and facial movements at delays of over 2 video frames (66 msec). Second, we have been investigating whether the rubber hand illusion first described by Botvinick & Cohen (1998) for visual-tactile information extends to visual-proprioceptive information. In our study, participants have a rubber hand attached to their own hand with a brace. They watch the rubber hand as they move their own hand behind a screen and then report on whether they experience the movement in the rubber hand. Our results to date indicate that the rubber hand illusion does extend to visual-proprioceptive integration.

T11) The Affect of Short Musical Sequences with Different Melodic Contours

Joshua Salmon and Bradley Frankland

This research examined the relationship between affect (emotional content) and the shape of 16 equi-temporal eight note musical sequences designed to differ only on melodic contour (ascending, descending, arch, inverted-arch, etc.). Two groups of participants rated the intended affect of each sequence on ten bipolar affect scales (contentment-joy, delicacy-strength, excitement-boredom, passivity-aggression, etc.). Modeling of melodic contour was done using fourth-order polynomial equations. Of which, the first term captures the average pitch height. The second term captures the ascending / descending aspect of the shape. The third term captures the arch aspect of the shape. Higher order terms capture nuances. The analysis indicated that pitch height was a primary predictor on eight of the affect scales. However, the analysis also indicated that the ascending/ descending aspect was important for five affect scales, while the degree of arch was important for eight affect scales. Different terms were important for different affect scales.

T12) Do the information processing differences in figure and ground reported by Wong & Weisstein (1983) generalize to a speeded orientation discrimination task?

Darren McKee

Wong and Weisstein (1983) reported that sharp line targets were detected better within a visual region perceived as figure while blurred line targets were detected better within a region perceived as background. They interpreted these results as evidence for two modes of visual processing: 1) high spatial-frequency channels that facilitate figure analysis and 2) low spatial-frequency channels that facilitate ground analysis. These results have not been replicated and until now, no published work

has demonstrated whether they generalize beyond the narrow range of methods used in the original study. The current investigation examined the generalizability of their findings using similar figure/ground and target stimuli and a better choice of target locations within the figure/ground drawing. The task was also changed from one of detection to one of horizontal/vertical line discrimination. The findings failed to replicate Wong and Weisstein. As such, their results do not generalize after minor methodological alterations.

T13) Mechanistic Issues in Human Coitus: Implications to Sexual Recovery after Iatrogenic Impotence

Richard Wassersug

Tens of thousands of men become impotent each year as a result of either the surgery or radiation used to treat prostate cancer. For approximately a third of these men, phosphodiesterase 5 (PDE5) inhibitors, such as Viagra, are ineffective. Many studies indicate that this loss of sexual function undermines the men's self-esteem and lowers their overall quality of life.

In this talk I will review options for these patients. I will discuss the limitations of current treatments for erectile dysfunction and introduce a counter-intuitive work-around for recovery of rewarding sex that completely circumvents the issue of penile erectility. I will provide a hypothesis, based on neural activity in cortical sensory fields, for why this approach works. I will also discuss the psycho-social challenges to making novel approaches to sex therapy acceptable to men and their partners, the majority of whom are senior citizens.

T14) The Role of Child Life in Pediatric Pain Management

Nancy Bandstra, L. Skinner, C. LeBlanc, C. Moores, D. Brennan, and C. Chambers

There is now strong support in the use of non-pharmacological approaches for reducing children's pain and distress during painful medical procedures. Although an exponential growth in knowledge in these techniques has occurred in the past decade, studies continue to suggest that children are undertreated for their pain. This research project is being conducted to examine child life specialists' role in pediatric pain management. While it is generally known that child life specialists make an important contribution in terms of pediatric pain management, no research has described the specific pain management strategies used by child life specialists. The purpose of the current study is 1) to describe the strategies currently being used by child life specialists to manage pediatric pain 2) to describe how effective child life specialists find these strategies to be, and finally, 3) to determine how much training experience child life specialists have in pediatric pain management and assess the level of interest in receiving more training. Child life specialists from hospitals across Canada will be invited to complete an online questionnaire concerning their use of various non-pharmacological pain management strategies to help pediatric patients cope with pain.

T15) Investigations of Affective Processing and Psychopathy using fMRI

Antisocial Personality Disorder (APSD) is a personality disorder characterized by a major affective deficit accompanied by a disregard for societal rules in general (e.g., Hare, 1996). A subset of those with APSD can be characterized as manipulative, callous, remorseless, irresponsible individuals (e.g., Hare, 1996, 1998), or as having psychopathic traits. According to Patrick (1994), a central feature of psychopathy is the deviant affective response. Since individuals with psychopathic characteristics appear to have emotional deficits, the question remains as to whether or not these deficits have a neurological basis. The limited existing studies that have used neuroimaging techniques implicate brain abnormalities in psychopathy and antisocial behaviour. The main objective of this program of research will be to determine how the brain processes affective visual and audio information in normal controls and in individuals with psychopathic characteristics. This research will shed light on the underlying neural deficits postulated to be associated with psychopathy, which may in turn help explain the antisocial behaviour that is characteristic of this disorder.

T16) Gender differences in implicit and explicit cognitions pertaining to anxiety

Kerry Lynn MacSwain and Margo C. Watt

Previous research has found that women (vs. men) yield higher anxiety scores on both explicit and implicit measures of cognitions related to trait anxiety and anxiety sensitivity (AS). In addition to this, the correlation between implicit and explicit measures of trait anxiety is higher for women, suggesting that explicit measures reflect this underlying personality construct more accurately in women than in men. The present study proposes to further investigate these gender differences in cognitions related to anxiety. Participants will complete two explicit measures and two implicit measures of trait anxiety and AS, including a newer implicit measure known as the Extrinsic Affective Simon Task (EAST). It is hypothesized that women (vs. men) will yield higher anxiety scores on implicit and explicit measures of cognition. It is also expected that the correlations between implicit and explicit anxiety measures will be higher for women than men.

T17) Sleep on it

Jennifer Vriend, Jason Ivanoff, Angeline Vaughan, John Christie, and Ben Rusak

The Visual Discrimination Task (VDT) is a sensory learning task, performance on which has been reported to depend on sleep intervening between initial acquisition and later testing. We studied VDT performance in 21 subjects (19 F, 2 M, aged 20-27), each in two conditions: once with, and once without, a 2 h nap opportunity (13:30-15:30) intervening between morning training (08:30) and evening testing (19:00). Performance improvement was assessed as the difference between morning and evening performance during each session. Subjects were divided into good versus bad nappers based on whether they showed at least 2 min of REM and 2 min of Slow Wave Sleep during the nap opportunity. Good nappers improved more during the day in the nap session than did bad nappers. This improvement, however, depended entirely on good nappers performing poorly during the morning of the nap session. The results suggest the hypothesis that sleep may improve newly acquired cognitive skills only in people who are relatively sleep deprived.

T18) The Effects of Positive and Negative Mood States on Mood-Regulation Gambling Outcome Expectancies

Jane Baldwin and Chris Mushquash

This study aimed to clarify how positive and negative mood states influence the mood-regulation gambling outcome expectancies of undergraduate gamblers, as self-reported on a recently developed questionnaire (the Gambling Expectancies Questionnaire; GEQ). Participants were 91 undergraduate psychology students, aged 18 or older, who reported regular gambling. Participants were randomly assigned to a positive or negative musical mood induction procedure (MMIP) or to a neutral mood control. The GEQ was administered at baseline and after the MMIP to determine changes in reward and relief expectancy strength. The hypotheses of mood specificity of the activation of reward and relief expectancies were only partially supported. The trend of increased reward expectancies following positive mood induction was expected; however, the trends of increased relief expectancies

following positive mood induction and decreased reward expectancies following negative mood induction were unexpected. Further research and potential clinical implications are discussed.

T19) The Effect of Outcome Expectancy on Depression and Quality of Life in Rural Stroke Survivors

Shannon Currie

Patient recovery expectations are related to health outcomes and have been found to be influence the success of treatment. The relationship between outcome expectancy and overall functioning has potential implications for the rehabilitative process of patients with a chronic illness; however this relationship has not yet been explored in stroke survivors. The purpose of the present study is to examine the relationship between outcome expectancies and depression and quality of life in rural stroke survivors. A measure of outcome expectancy was designed to assess 13 different domains of outcome within 6 broad categories of memory, physical functioning, social relationships, leisure and occupational activities, dependence, and overall expectations. This measure was included in the Yarmouth Stroke Project (YSP) interviews, which are currently being conducted and include qualitative and quantitative questions that address psychosocial functioning. The development and efficacy of the expectancy measure will be discussed with reference to preliminary results if available.

T20) Are my eyes playing tricks on me? Evidence of inhibitory neurotransmitter actions of β -alanine in the salamander retina

Alana R. Paul and Steve Barnes

β -Alanine is a putative inhibitory neurotransmitter in the central nervous system. Using calcium imaging, the responses of neurons in living retinal slices were investigated during exposure to β -alanine, glycine, GABA_A agonist, muscimol, and GABA_C agonist, cis-4-aminocrotonic acid. At 1 mM, β -alanine caused inhibition of depolarization-induced calcium responses in both photoreceptors (PRs) and inner nuclear layer (INL) cells, similar to results seen with 500 μ M muscimol and glycine. However, 200 μ M β -alanine caused enhancement (possibly due to disinhibition) of calcium response in PRs, and an ambivalent response in INL cells. Neither glycine nor muscimol caused this enhancement. It is possible that β -alanine can inhibit both PRs and INL cells, but that it has more potent effects in the INL and thus at lower concentrations, β -alanine selectively inhibits cells in the INL. This response to β -alanine could provide evidence for a novel β -alaninergic signalling pathway in the retina involving inhibition of cells presynaptic to PRs.

T21) Non-Phosphorylated Neurofilament Labeling is Regulated by Visual Experience in the Kitten Lateral Geniculate Nucleus: Implications for Anatomical Remodeling During the Critical Period.

Matthew Kutcher and Kevin Duffy

Monocular deprivation (MD) during early development causes rearrangement of neural connectivity that can produce severe and permanent vision impairment. Neurons in deprived layers of the lateral geniculate nucleus (LGN) exhibit smaller somata and reduced geniculocortical terminal fields compared to neurons in non-deprived layers. The mechanisms underlying these deprivation-induced changes are unknown. We have demonstrated a correlation between structural change and cytoskeleton alteration in visually deprived cats. Following 4 days of MD, neurons within deprived layers of the LGN were smaller than those in non-deprived layers, and this structural change was accompanied by a spatially and temporally coincident reduction in neurofilament labeling. Recovery of cell-size and neurofilament labeling was observed after 4 days of post-deprivation binocular visual experience. The recovery appeared to be complete after 8 days of binocular vision. The results from our study support the hypothesis that cytoskeleton alteration provides a substrate for deprivation-induced neuronal restructuring.

T22) Not all visual input is created equal: Normal input is more efficacious for development of the visual cortex and for vision.

Donald E. Mitchell and Frank Sengpiel

Continuous monocular deprivation (MD) during early postnatal life results in reduced cortical representation and visual acuity of the deprived eye. However, loss of visual function in one eye after a brief period of compromised vision would appear to be maladaptive. Previously we have reared cats with a daily regimen in which they received variable amounts of MD (by wearing a mask) and BE, and subsequently assessed the visual capabilities of the two eyes. Here we report on visual cortex function on the basis of optical imaging of intrinsic signals. As with behaviour, we show that daily amounts of binocular vision as short as 0.5 h allowed normal ocular dominance architecture and visual responses to develop despite much longer periods of MD. However, unlike vision where the outcome appears to be dictated by the proportion of total exposure that is normal, cortical architecture appears to be determined by the absolute amount of binocular vision rather than its relative share. As with behaviour, continuous binocular experience was more effective for cortical development than an equivalent exposure split into two epochs. Normal experience is clearly more efficacious in driving visual cortical development and vision than abnormal experience.

T23) The Swim of the Mummichog

Heather-Dawn Wood, Jacob Lingley and Simon Gadbois

The present experiment sought to determine if *Fundulus heteroclitus* (the common killifish or mummichog) would use distal visual cues or proximal landmarks to complete an innovative place preference paradigm. A plus maze consisting of four empty beige saucers and one enriched darker saucer was used to exploit the mummichog's innate preference for concealment. In condition 1, 10 fish were trained with a high-contrast, extra-maze distal cue, situated over the enriched saucer. In condition 2, 10 fish were trained with proximal landmarks situated in the entrance of the arm leading to the enriched saucer. Experimental trials consisted of displacing the visual cues and recording accuracy. Error rate for condition 2 was found to be significantly lower than condition 1, indicating fish using proximal landmarks had significantly fewer errors than fish using distal cues. These results promote the present paradigm as a useful fish cognitive task for research across the fields of ecotoxicology, ethology, and psychology.

T24) Learning about Learning from Zebrafish

Oliver Braubach, Alan Fine and Roger Croll

The zebrafish olfactory system is emerging as an important model for studying neural processing of chemosensory information. This is largely due to its accessibility for *in vivo* physiological studies and also the ease of genetically manipulating this system. While our knowledge about the neurophysiology of zebrafish olfaction is advancing rapidly, these fish have not yet been widely used in studies of olfactory behavior. We therefore sought to develop various assays to study olfactory-driven behaviors and learning in zebrafish. We have succeeded in conditioning zebrafish to respond to specific odorants in order to obtain a food reward, while at the same time conditioning them to ignore other odors of similar quality. Physiological recordings in the zebrafish olfactory system will enable us to pinpoint regions in the olfactory bulb that are responsive to specific odorants. Combining these approaches allows us to induce behavioral plasticity that can be correlated with detailed physiological data to reveal neural substrates activated and modified in olfactory learning.

T25) Sleep on it

Jennifer Vriend, Jason Ivanoff, Angeline Vaughan, John Christie, and Ben Rusak

The Visual Discrimination Task (VDT) is a sensory learning task, performance on which has been reported to depend on sleep intervening between initial acquisition and later testing. We studied VDT performance in 21 subjects (19 F, 2 M, aged 20-27), each in two conditions: once with, and once without, a 2 h nap opportunity (13:30-15:30) intervening between morning training (08:30) and evening testing (19:00). Performance improvement was assessed as the difference between morning and evening performance during each session. Subjects were divided into good versus bad nappers based on whether they showed at least 2 min of REM and 2 min of Slow Wave Sleep during the nap opportunity. Good nappers improved more during the day in the nap session than did bad nappers. This improvement, however, depended entirely on good nappers performing poorly during the morning of the nap session. The results suggest the hypothesis that sleep may improve newly acquired cognitive skills only in people who are relatively sleep deprived.

T26) “I Sleep Therefore Can I Think”: Effects of Obstructive Sleep Apnea on Neurocognitive Function

Esther Y Y Lau

Obstructive sleep apnea (OSA) is characterized by disrupted breathing during sleep, leading to hypoxemia and fragmented sleep at night, daytime sleepiness, and changes in cognition and mood. Individuals treated with CPAP (continuous positive airway pressure) report improved fatigue and better sleep, yet commonly continue to have complaints of persistent cognitive dysfunction and reduced performance at work and daily activities, particularly those that require executive functions.

The purpose of the study is to investigate the executive function of individuals with OSA on treatment with CPAP. OSA patients and matched controls will be compared on experimental working memory tasks, standardized neuropsychological battery, measures of mood, daily functioning and quality of life. All patients and some controls will also receive an overnight polysomnographic sleep study to characterize their current sleep. It is hypothesized that while CPAP may improve sleep and thus attentional capacity related to the excessive daytime sleepiness, performance on tasks measuring executive control will continue to be impaired.

T27) The Effect of Mood on Susceptibility to Misinformation Regarding Emotional Photographs

Sabrina Bellhouse

The current study will investigate the influence of mood and the emotional valence of stimuli on accuracy of recall and susceptibility to the misinformation effect. A negative, positive, or neutral mood will be induced in participants using a musical mood induction technique. Participants will then be shown three emotional photographs: one negative, one neutral, and one positive, and after a delay will be asked accurate or misleading questions regarding details contained in the photographs. After a second delay, half of the participants will answer open-ended questions about certain details that were in the photographs that they saw, and the other half will be shown two photographs (one original and one altered to incorporate the misinformation in the misleading questions) and will be asked to select the original photograph that they saw.

T28) Differences in High and Low Anxiety Sensitive Women's Responses to a Laboratory-Based Cold Pressor Task

Lindsay S. Uman, Sherry H. Stewart, Margo C. Watt and Amber Johnston

Anxiety sensitivity (AS) or the fear of anxiety-related sensations has been recently linked to a variety of disabling chronic health conditions including pain-related conditions (Asmundson et al., 2000). In the present study, 90 undergraduate university women were selected for inclusion in one of two AS groups (high or low) based on their screening scores on a 16-item measure of AS. Participants were tested individually on a lab-based cold pressor task using a variety of self-report and observer-measured variables. Data analyses revealed that, as expected, the high AS participants reported significantly more fear in response to the cold pressor on the McGill Pain Questionnaire – Short Form (SF-MPQ) than did the low AS participants. Also as expected, the high AS participants reported more pain in response to the cold pressor on the Present Pain Index (PPI) of the SF-MPQ than did the low AS participants. High AS participants did not differ from low AS participants on other aspects of the cold pressor response (e.g., pain threshold, pain tolerance, pain recovery). Results also provide support for the role of pain-related fear as a mediating variable between AS and increased perceived pain intensity.

T29) ?I didn?t forget it, I just remembered something else instead!?: Examining possible attentional allocation in directed forgetting.

Angela Arsenault

A standard item-method directed forgetting task presented sequential word pairs consisting of two differently colored words. Participants were instructed to attend or ignore designated words based on color. Immediately following the presentation of each word pair, a remember or forget memory cue was presented aurally. Following study, a recognition test assessed memory of both forget and remember-cued words, regardless of attention instruction (attend or ignore). As expected, percentage correct scores for attended words illustrate superior recognition performance on remember-cued as compared to forget-cued words, indicative of a robust directed forgetting effect. In contrast, memory performance on ignored words shows an interesting potential trend for a reverse directed forgetting effect. It appears as though withdrawal of attention from forget-cued words is succeeded by attention to irrelevant information, such as ignored words. Support for attention models of directed forgetting and applications will be discussed.

T30) The Relationship between Attention and Forgetting in Humans

Jonathan Fawcett and Tracy L. Taylor

Reaction time (RT) was measured in response to visual detection probes embedded within an item-method directed forgetting paradigm. Study words were presented individually followed by an instruction to remember (R) or forget (F). Probes were presented at stimulus onset asynchronies (SOAs) of 1400, 1800, or 2600 ms in relation to the study word or the memory instruction. No-probe catch trials and a no-word control condition were included to evaluate false alarms and provide a subtractive baseline measure. At the end of all study trials, a yes-no recognition task measured retention of R and F words. The results demonstrated a tendency for post-F instruction probe RTs to be longer than post-R instruction probe RTs at the two earliest SOAs; this effect was marginally significant at the 1400 ms SOA and statistically significant at the 1800 ms SOA. Recognition data indicated a significant directed forgetting effect, with better recognition of R than F words. These findings suggest that intentional forgetting requires more attentional resources than remembering.

T31) Learning and memory in C57BL/6J mice on the Barnes maze with a fixed or randomly located escape hole.

Timothy P. O'Leary and Richard E. Brown

The Barnes circular maze is a visuo-spatial learning and memory task that was originally designed for rats, and has been adapted for use with mice. In the Barnes maze, animals use extra-maze visual cues to locate a hole that provides escape from aversive bright light. To determine if mice use extra-maze visual cues in the Barnes maze, male and female C57BL/6J mice were trained with the escape hole in a fixed location, or moved randomly across trials, so that extra-maze visual cues could not predict the location of the escape hole. Mice trained with the escape hole in a fixed position made

fewer errors, traveled shorter distances and searched more accurately during probe trials than mice trained with a randomly positioned escape hole, indicating that mice may use extra-maze visual cues in the Barnes maze. Sex differences were found for some training phases but not others, and will be discussed.

T32) The Effects of Repeated Exposure to Cat Odour on Behavioural and Endocrinological Stress Responding in the Rat

Lisa Wright, Rahia Mashoodh, Kimberly Hébert and Tara Perrot-Sinal

Acute exposure of a rodent to a predatory odour is an ethologically appropriate stressor that induces robust and quantifiable behavioural and physiological responses. Predator odour (PO) exposure therefore models psychological stress experience and may be used to study the long-term consequences of continued or repeated activation of the hypothalamic-pituitary-adrenal (HPA) axis and hypersecretion of glucocorticoids (GCs), which are linked to the development of various anxiety and depression-related neuropsychiatric illnesses in both men and women. The goal of the present study was to characterize changes in behavioural and hormonal responses to predation threat across a period of chronic odour exposure, and also to examine sex differences in such changes. This was achieved by exposing groups ($n=12$) of adult male and female Long Evans rats to a PO or control odour (CO) stimulus for one hour/day over a period of 22 days. Behavioural responses were examined on the first exposure day and every fourth day thereafter. Corticosterone (CORT) levels were assayed from blood samples collected during a baseline condition, and following exposure conditions at the beginning and end of the chronic odour exposure period. Interestingly, patterns of behavioural change across the exposure period were similar for PO and CO exposed rats, although the magnitude of behaviours remained very different. There were also sex-specific alterations in GC release, with both CO and PO exposed females showing higher levels of plasma CORT at the end of the chronic odour exposure period relative to the beginning. In a subsequent study, we will characterize the temporal profile of GC release following initial exposure to PO.

T33) Flavour-Flavour Learning in a Devaluation Procedure Controlling for Post-Ingestive Effects.

Gillian M. Alcolado & Vincent M. LoLordo

We were looking for evidence of flavour-flavour learning across long delays between flavour presentations. Twenty-six male rats received pairings of a neutral flavour with a delayed, differently flavoured carbohydrate, and pairings of a second neutral flavour with another delayed, differently flavoured carbohydrate. A third neutral flavour was presented alone. Then one of the flavoured carbohydrates was devalued by pairing it with an injection of lithium chloride. All rats avoided the devalued flavoured carbohydrate in a test pitting it against the non-devalued flavoured carbohydrate, supporting the effectiveness of the devaluation procedure. Most rats failed to avoid the neutral flavour that had been paired with the devalued carbohydrate when it was pitted against the neutral flavour that had been paired with the non-devalued carbohydrate. However, one subgroup that didn't have mint as one of the flavours paired with a carbohydrate learned the association, suggesting that replication with truly neutral flavours is merited.

T34) Utilizing Caffeine Dependence To Condition An Avoidance To A Novel Flavor

Sarah Boudreau

Male Sprague-Dawley rats were given chronic exposure to caffeine in their drinking water for 21 days in an effort to establish caffeine dependence. They were then given daily 30-minute trials with a Kool-Aid/Saccharin flavor (Maintenance CS), after each of which the caffeinated drinking water was returned. Rats were then given daily 30-minute access to a different Kool-Aid/Saccharin flavor (Withdrawal CS), followed by tap water allowing the rats to go into caffeine withdrawal. All rats were then given a 24-hour two-bottle test pitting the Maintenance and Withdrawal CSs against each other. The rats overwhelmingly avoided the Kool-Aid/Saccharin flavor paired with caffeine withdrawal symptoms, and drank more of the Maintenance CS. Based on these results, we can conclude that caffeine dependence was established, and that dependence can be used to condition an avoidance to a flavor paired with caffeine withdrawal.

POSTERS

P1) Expressive and Receptive Pain Language and Its Relationship to General Language Development Among Healthy Children

Tessa J. Craig, and Christine T. Chambers

Recent research has shown that young children tend to rely on a select number of words to describe pain. The purpose of the current study was to extend previous research by examining receptive as well as expressive pain language in children, and to examine the relationship between pain language and children's general vocabulary skills. Sixty children (30 boys; 30 girls) aged 3-6 years ($M = 4.17$, $SD = 1.03$) and a parent (93% mothers) participated in the study. Parents completed a questionnaire assessing the degree with which their children used and understood seven words for pain, and the age at which they first started to use and understand these words. Children's general vocabularies were assessed using two standardized measures of vocabulary development: the PPVT-III and the EVT. Results indicated that 'hurt' was used most frequently and was best understood by the children, appearing in their vocabularies at 28 months of age. 'Pain' and 'ache' were rated as being infrequently used and poorly understood, and did not appear in children's vocabularies until around 40 months of age. There were no significant relationships between children's expressive and receptive pain language and their scores on the measures of general expressive and receptive vocabulary (r 's = .06 to .22). These results have implications for developmentally appropriate assessment of pain in young children.

P2) Equating Secondary Tasks: A Dual-Task Investigation in 5-Year-Old Children Or... Doing Two Tasks at the Same Time Should be Harder Than Doing Just One (But Apparently It Isn't)

Jennifer A. Parker and Sophie Jacques

Two experiments were performed with 5-year-old children to establish equivalent verbal and nonverbal secondary tasks for use in future dual-task studies. Children's response time on a simple primary task was measured while they were asked to perform a series of concurrent secondary tasks. The primary task in the first experiment required children to touch a target on a computer screen, while in the second experiment, they were asked to draw a continuous path. Children in both studies participated in five secondary task conditions. They were asked to repeat one word continuously in two verbal secondary tasks (i.e., "ball" vs. "candy"); they were asked to make a facial movement in the two nonverbal secondary tasks (i.e., smack their lips vs. make a kissy face); and they were not asked to perform any secondary tasks in a control condition. Interestingly, it was found that children's response times in the control condition were not faster than those in secondary task conditions. These findings are consistent with Luria's theory on the verbal regulation of motor behaviour.

P3) Using the Attention Network Test to Measure Three Types of Attention in Children with Attention Deficit/Hyperactivity Disorder

Meredith Pike, Jennifer Mullane and Penny Corkum

The Attention Network Theory (Posner & Peterson, 1990) proposes that there are three functionally and anatomically independent networks of attention: Executive, Alerting and Orienting. The Executive and Alerting networks are thought by some researchers to be impaired in children with ADHD. Recently, the Attention Network Test (ANT) was developed to measure these three networks simultaneously. To date, the ANT has never been used in children with ADHD. In the current study, the ANT was given to 20 children with ADHD. The results validate the utility of the ANT as a new measure for studying attention in children with ADHD. As well, further evidence for the independence of the three attention networks was found which provides further evidence of support for the Attention Network Theory.

P4) The Influence of Subtype on Academic Difficulties and Cognitive Profiles in Children with ADHD

Laura Knickle, Melissa McGonnell and Penny Corkum

It has been proposed that the subtypes of ADHD differ in their cognitive and academic profiles, as well as in their symptom profiles and demographic characteristics. However, the results of previous research has been inconsistent, most likely due to a number of limitations, including poorly diagnosed ADHD samples and lack of standardized assessment of cognitive and academic skills. The present study included 64 children who received a diagnosis of ADHD-Combined Type or ADHD-Inattentive Type based on a rigorous, interdisciplinary assessment process. It was found that the ADHD subtypes did not differ in terms of their age of diagnosis, sex distribution, mean IQ scores or academic performance, but that the ADHD-Inattentive Type had less comorbid mental health problems and higher socio-economic level. The results question the clinical utility of the subtypes in helping to understand cognitive and academic performance in elementary school-aged children.

P5) Impact of Methylphenidate on Sleep and Daytime Alertness in Children with Attention-Deficit/Hyperactivity Disorder.

Sarah Ironside and Penny Corkum

Changes in sleep quality related to treatment with methylphenidate and potential daytime sequelae of sleep disturbances were evaluated across a randomized placebo-controlled medication trial in 15 children, rigorously diagnosed with attention-deficit/hyperactivity disorder (ADHD). Data were analyzed with a repeated measures ANOVA and contrast analyses. Parents and teachers reported an improvement in ADHD symptoms on medication. Parent questionnaires and actigraphy indicated increased sleep disruption on medication. Significant changes in sleep duration, time to fall asleep, and wake episodes were recorded, with no significant changes in sleep efficiency or nocturnal activity. No changes in daytime alertness were reported, but preliminary evidence of hypersomnolence after discontinuation of medication is presented. Results and implications of the study for treatment of children with ADHD are discussed.

P6) Effects of sleep deprivation on hippocampal plasticity in adult rats

Adrienne Junek, Ben Rusak and Kazue Semba

Sleep deprivation for 96 h decreases hippocampal cell proliferation in rats (J Physiol 2003 549:563). We are currently investigating whether sleep deprivation for 6, 12, 24, and 48 h can produce the same effect. We are also investigating any changes in hippocampal PSA-NCAM expression after sleep deprivation. Adult male rats were sleep deprived using a modified Wahmann running wheel, injected with bromodeoxyuridine (BrdU, a marker for cell proliferation) 2 h before the end of the SD, and perfused at the end of SD. Blood samples were also collected for plasma corticosterone analysis in order to assess the stressfulness of SD. Preliminary counts of BrdU-immunoreactive cells in the dentate gyrus show a small but significant increase in cell proliferation after 12 h SD, and a trend for decreased cell proliferation after 24 and 48 h SD.

P7) Event-related potential methods for evaluating functional status in epilepsy

Erin L. Mazerolle, Ryan C.N. D'Arcy, and R. Bruce Bolster

Neuropsychological assessment in temporal lobe epilepsy (TLE) can be augmented by event-related potential (ERP) methods that evaluate the functional status of tissue adjacent to medial epileptogenic foci. Anterolateral temporal regions have been linked to visual object recognition (VOR) in healthy controls. We propose an ERP method for assessing the functional integrity of anterolateral temporal regions using a word-picture matching task. The semantic level of the word (basic versus superordinate) and the object category (natural versus artifactual) were manipulated in order to identify components that are sensitive to intact VOR. Distributed sources were modeled to determine whether these components were localized to anterolateral temporal lobe regions. ERP components that were sensitive to both semantic level and object category manipulations were identified in healthy controls. Furthermore, one of these components was localized to the anterolateral temporal lobes.

The findings demonstrate that the spatiotemporal information provided by ERPs can potentially be used to evaluate the functional status of the anterolateral temporal lobes. This approach can be applied in epilepsy assessment to, for example, guide surgical intervention in TLE.

P8) Spiraling into Control: Overcoming fMRI susceptibility to artifacts in the inferior temporal lobe

Kim Brewer, Jodie Gawryluk, James Rioux, Chris Bowen, Ryan C.N. D'Arcy and Steven Beyea

We are developing a site directed functional magnetic resonance imaging (fMRI) approach to guide pre-surgical mapping in patients with temporal lobe epilepsy (TLE). Susceptibility effects are the basis of functional imaging, however, susceptibility induced signal loss also occurs at numerous tissue-sinus interfaces. This signal loss creates artifacts in the inferior temporal lobes and orbitofrontal area, thereby limiting the ability to detect functional activation in these regions. The current investigation utilized a 4 tesla imaging system to examine alternative fMRI techniques with the objective of reducing artifacts caused by susceptibility. Specifically, the commonly utilized technique of echo-planar imaging (EPI) was compared to spiral out and spiral in methods of image acquisition. Compared to EPI and spiral out, the spiral in technique has produced promising results. Utilizing spiral in, it is possible to gain functional information at the tissue-sinus interface. To our knowledge, this represents the first demonstration of spiral in methods at 4 tesla. Application of this novel technique will allow for advanced pre-surgical functional mapping research on temporal lobe epilepsy.